

Solving Word Problems Involving Unit Conversions

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Unit conversion is used to express the same measurement using a different unit.

Step 1: Identify

Read the word problem carefully and identify:

- ▶ the **given**
- ▶ the **required**
- ▶ the **unit** needed in the final answer

Step 2: Use a Conversion Factor

Use the appropriate **conversion factor** to change one unit into another.

Example factor: $1 \text{ km} = 1000 \text{ m}$

Step 3: Cancel Unwanted Units

Make sure unwanted units **cancel out** during the conversion process.

$$3 \cancel{\text{km}} \times \frac{1000 \text{ m}}{1 \cancel{\text{km}}} = 3000 \text{ m}$$

Step 4: Label Your Answer

Always write the **correct unit** with your final answer.

An answer without a unit is incomplete!

Examples

Example 1

Jake ran 3.2 km around the track. How many meters did he run?

Given: 3.2 km **Required:** meters

$$3.2 \times 1000 = 3200 \text{ m}$$

Example 2

A jug holds 2.4 L of water. Express the amount in milliliters.

Given: 2.4 L **Required:** milliliters

$$2.4 \times 1000 = 2400 \text{ mL}$$

Example 3

A bag of sugar weighs 3 kg. If sugar costs ₱60 per kilogram, how much does the bag cost?

Given: 3 kg at ₱60/kg **Required:** total cost

$$3 \times 60 = \text{₱}180$$

Example 4

A road is 7 000 m long. How many kilometers is that?

Given: 7 000 m **Required:** kilometers

$$7\,000 \div 1\,000 = 7 \text{ km}$$

Practice Exercises

Practice 1

Maria walked 2.8 km from her house to school.
How many meters did she walk?

Practice 2

A bottle contains 3.5 L of juice. Express the amount in milliliters.

Practice 3

A carpenter needs a wooden plank that is 250 cm long. Express its length in meters.

Practice 4

A motorcycle traveled 18 000 m. Express the distance in kilometers.

Practice 5

A baby weighs 4.2 kg. Express the weight in grams.

Practice 6

A water tank holds 7 500 mL of water. Express the capacity in liters.

Practice 7

A recipe requires 1.5 kg of flour. If flour costs ₱80 per kilogram, how much will the flour cost?

Practice 8

A runner completed a race of 5 000 m. How many kilometers did the runner complete?

Activities

Activity 1

Carlo biked 4.6 km around the park. How many meters did he travel?

Activity 2

A container holds 2.8 L of milk. Express the amount in milliliters.

Activity 3

A ribbon measures 360 cm. Express its length in meters.

Activity 4

A train traveled 42 000 m. Express the distance in kilometers.

Activity 5

A sack of rice weighs 25 kg. Express the weight in grams.

Activity 6

A pitcher contains 9 000 mL of water. Express the capacity in liters.

Activity 7

A fruit vendor bought 2.5 kg of apples. If apples cost ₱120 per kilogram, how much did the vendor pay?

Activity 8

A delivery truck traveled 86 km. Express the distance in meters, then determine how many meters the truck traveled.

Activity 9

A container holds 3.6 L of fruit juice. It was poured equally into 8 bottles. How many milliliters of juice were placed in each bottle?

Activity 10

A cyclist completed a route measuring 12 000 m. How many kilometers did the cyclist travel?

Thank you!